Hsuan-Fu (Jeff) Hsu

Embedded Systems and AI Engineering | C++, Raspberry Pi, Arduino, Python

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EDUCATION

University of Southern California — Los Angeles, CA
Master of Science in Computer Science | GPA: 3.6/4.0 — Jan 2026 - May 2027 (Expected)

National Taipei University — New Taipei City, Taiwan
Bachelor of Science in Computer Science | GPA: 3.5/4.0 — Sept 2020 - June 2024

SKILLS

Languages: C++, Python, C#, TypeScript, SQL

Embedded and Graphics: Raspberry Pi 4, Arduino Nano 33 BLE Sense, I2C, BLE, OpenGL, GLSL

ML and Data: PyTorch, TensorFlow, Sentence Transformers, NumPy, tsfresh

Engineering: Git, Linux, Docker, AWS (Lambda, SQS, DynamoDB), pytest, xUnit, GitHub Actions

EXPERIENCE

National Taipei University — New Taipei City, Taiwan
Undergraduate Researcher | Advisor: Prof. Bor-Shing Lin — July 2022 - Feb 2024

- Co-designed and built a portable NIR biosensing device with two other core contributors using **Arduino Nano 33 BLE Sense**, an AS7263 sensor over I2C, BLE transmission, and a 3D-printed enclosure.
- Built a **Python** pipeline for noninvasive uric acid estimation from 30 participants and two wavelengths; the team's linear regression model achieved 10.3% MAPE with 10 selected features under LOSO evaluation.
- Received individual NSTC undergraduate research funding and earned fourth place in NTPU's project competition with a team of three.

Tech-Management Workshop Co., Ltd. — Taichung, Taiwan
Software Engineer Intern — July 2026 - Aug 2026

- Independently built a government birth grant application with **ASP.NET Core MVC** and **SQL Server**, deployed across 29 offices with application status tracking.
- Implemented incremental synchronization across the offices, replacing manual uploads of complete datasets with transfers of changed records.
- Reduced database latency by at least 50% and overall resource usage by at least 30% through data processing and storage improvements.

PROJECTS

Classroom AC Control System — USC IEEE HackIoT (team of 4) — Oct 2026
Raspberry Pi 4, Python, YOLO, DHT11, Supabase, TypeScript | GitHub

- Set up a **Raspberry Pi 4** with teammates to read a DHT11 temperature and humidity sensor twice a minute and count people with a Logitech camera running YOLO at about 7 frames per second, then sent the readings to Supabase.
- Ported a teammate's Python AC setpoint model to **TypeScript**, matched it hour by hour over 1,344 hours, and built a Next.js simulator; in a simulated 8-week run, one study room used an estimated 27% less energy.

3D Rendering Engine — Sept 2023 - Dec 2023
C++, OpenGL, GLSL | GitHub

- Built a custom **C++** OBJ/MTL parser and **OpenGL** rendering pipeline using VAOs/VBOs and model, view, and projection transforms.
- Implemented texture mapping, multiple-light Phong shading, skybox rendering, and environment mapping with **GLSL** shaders.

PUBLICATION

Coauthor, "Noninvasive Portable System With Dual Near-Infrared Light Technology for Monitoring Uric Acid Concentration," IEEE Transactions on Human-Machine Systems, accepted for publication, Aug 2026.